ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Teaching and answer guidance · Milling and reclaiming

Volume 5 · H1, H2, H3, H4, H5, H6, H7, H8

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Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for milling and reclaiming. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS
Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT
Audio file / video container / timed web captions; test playback and retain a transcript.

Milling and reclaiming equipment

Worked example · A noise with a location

A new rhythmic noise and vibration begin at the discharge conveyor after a load change. A learner suggests increasing speed to clear it.

Report the change and location, stop safely under the machine procedure and protect the area. Have the responsible person arrange inspection under the required isolation. Increasing speed is an unsupported intervention and may worsen the fault.

Match rotor, discharge conveyor, lifting column and grade sensor to their functions.

Match rotor, discharge conveyor, lifting column and grade sensor to their functions.

Answer guidance.

Rotor cuts; discharge conveyor transfers recovered material; lifting column changes/supports machine elevation as designed; grade sensor reads a reference for control. Identify exact configurations on the selected machine.

Observe: Reading, Communication through the learner's actual contribution, calculation, explanation or saved work.

A 90 m × 2 m strip is milled 0.04 m deep

A 90 m × 2 m strip is milled 0.04 m deep. Find its geometric removed volume. Is this its loose truck volume?

Answer guidance.

90 × 2 × 0.04 = 7.2 m³ in place. Loose volume requires a supported swell/conversion factor; it is not automatically 7.2 m³.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Make a digital component card with model, image, name, function and hazard.

Make a digital component card with model, image, name, function and hazard.

Answer guidance.

A valid card retains source/manual page and model scope, identifies one visible component accurately and explains its moving/energy hazard without inventing operating settings.

Observe: Digital, Writing, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify ten components and explain their functions.	0 / 1 / 2	Evidence/help: _____
Distinguish milling, reclaiming and stabilized-base processes.	0 / 1 / 2	Evidence/help: _____
Record a fault and the appropriate hold/escalation.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Milling and reclaiming safety procedures

Worked example · A changed discharge route

The planned truck position is blocked by a delivery vehicle. Someone proposes swinging the conveyor over a pedestrian route.

Stop the conflicting movements, preserve the pedestrian boundary and ask the lead to revise the arrangement with traffic control. Check overhead and swept-path hazards. Resume only after the new plan is communicated and controls are in place.

Name four hazardous energies that can remain after engine shutdown.

Name four hazardous energies that can remain after engine shutdown.

Answer guidance.

Gravity in raised components, hydraulic pressure, rotating inertia and electrical/stored energy may remain. The actual machine assessment determines all sources and controls.

Observe: Reading, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Write a read-back for a revised milling boundary at station 0+120.

Write a read-back for a revised milling boundary at station 0+120.

Answer guidance.

State station, lane/side, revised limit, discharge route, protected public route, stop signal and person authorizing the change. Ask the lead to confirm the interpretation.

Observe: Writing, Communication, Collaboration through the learner’s actual contribution, calculation, explanation or saved work.

A dust-control alarm appears while cutting

A dust-control alarm appears while cutting. What evidence should be retained?

Answer guidance.

Record machine, time, station, alarm and observed conditions; stop or hold as required by the safe procedure and notify the lead. Preserve the fault and correction record before authorized resumption.

Observe: Digital, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Brief and verify the work-zone boundary.	0 / 1 / 2	Evidence/help: _____
Demonstrate the approved lockout with all energy sources controlled.	0 / 1 / 2	Evidence/help: _____
Secure the parked machine and communicate its status.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Milling and reclaiming operator roles and responsibilities

Worked example · Two people signal

One ground worker signals a truck forward while another signals stop. The operator sees both.

Stop the affected movement. Clarify the hazard and designated signaller, confirm the driver and operator understand the revised instruction, and resume only when the area is clear and the communication arrangement is restored.

Assign operator, depth checker, truck coordinator and recorder duties for a simulated pass.

Assign operator, depth checker, truck coordinator and recorder duties for a simulated pass.

Answer guidance.

A workable allocation names each person, safe position, information needed, communication channel, stop authority and relief arrangement. Roles may be combined only where the plan and workload allow.

Observe: Collaboration, Writing through the learner’s actual contribution, calculation, explanation or saved work.

Depth readings at three authorized checks are 38, 41 and 40 mm against a fictional target 40 mm

Depth readings at three authorized checks are 38, 41 and 40 mm against a fictional target 40 mm. Find the mean and explain its limitation.

Answer guidance.

Mean = 119/3 = 39.67 mm. It does not prove every location meets the project tolerance; retain each reading and its location.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Rehearse a lost-radio response with a partner, then improve the wording.

Rehearse a lost-radio response with a partner, then improve the wording.

Answer guidance.

Stop affected movement, maintain safe positions, restore communication and confirm the instruction before restarting. An improvement uses machine ID, location and explicit acknowledgement.

Observe: Communication, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Perform a clear role and communication briefing.	0 / 1 / 2	Evidence/help: _____
Demonstrate stop on ambiguous or lost communication.	0 / 1 / 2	Evidence/help: _____
Report cut observations with location and units.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Milling and reclaiming equipment maintenance

Worked example · The wrong grease assumption

A chart specifies a particular grease and service interval. A worker brings an unlabelled cartridge and proposes ten strokes at every fitting.

Do not use the unlabelled product or generic stroke count. Confirm lubricant, points, quantities and safe positions from the matching chart. Obtain the correct product and record any blocked or damaged fitting for action.

A machine meter reads 812 h; the last service was at 786 h and the fictional interval is 25 h

A machine meter reads 812 h; the last service was at 786 h and the fictional interval is 25 h. Is it due?

Answer guidance.

812 − 786 = 26 h; it is 1 h beyond the stated exercise interval. Notify the lead and follow the actual maintenance/operating restrictions.

Observe: Numeracy, Reading through the learner’s actual contribution, calculation, explanation or saved work.

Write a useful entry for water found in a separator.

Write a useful entry for water found in a separator.

Answer guidance.

Record model/serial, meter, date, component, observed quantity/appearance, action taken under the procedure, waste handling and person notified. Do not diagnose the source without evidence.

Observe: Writing, Digital, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

A partner wants to check a leak with a bare hand

A partner wants to check a leak with a bare hand. Respond and arrange the next step.

Answer guidance.

Stop the unsafe approach, keep clear, secure the machine as required and refer inspection to authorized personnel using the approved method. Explain the injection hazard respectfully.

Observe: Communication, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Match lubricant/fluids and service intervals to the machine.	0 / 1 / 2	Evidence/help: _____
Service intake, separator and accessible points under the correct isolation.	0 / 1 / 2	Evidence/help: _____
Inspect cooling/battery areas, restore guards and complete the defect record.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Markers, grades and stakes

Worked example · Read an offset

A fictional stake says STA 0+125, 2.0 m RT OFFSET, FG EL 102.400, C 0.060. The legend defines cut from the marked reference to the required grade.

Locate station 125 m on the alignment and interpret the right-side offset using the project convention. The stated cut is 60 mm from the defined reference, not automatically from the top of the stake. FG is 102.400 m in the stated datum. Confirm the target location and mark before using the value.

Expand BM, CB, EL, FG, PL and PVC.

Expand BM, CB, EL, FG, PL and PVC.

Answer guidance.

Benchmark; catch basin; elevation; finished grade; property line; polyvinyl chloride. Verify project-specific usage.

Observe: Reading, Writing through the learner’s actual contribution, calculation, explanation or saved work.

At a verified point, existing EL is 102.455 m and required milled EL is 102.400 m

At a verified point, existing EL is 102.455 m and required milled EL is 102.400 m. Find nominal cut depth.

Answer guidance.

102.455 – 102.400 = 0.055 m = 55 mm, subject to the approved milling plan and checks for utilities and layer limits.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

A digital plan uses a geodetic datum; a loose note says “BM 100 assumed.” What should the crew do

A digital plan uses a geodetic datum; a loose note says “BM 100 assumed.” What should the crew do?

Answer guidance.

Do not combine them. Ask the survey/grade lead for the verified relationship and controlling reference. Record the clarification and communicate it to everyone using the marks.

Observe: Digital, Communication, Collaboration through the learner’s actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Interpret station, offset, cut/fill and elevation on a training layout.	0 / 1 / 2	Evidence/help: _____
Identify and protect benchmarks and monuments.	0 / 1 / 2	Evidence/help: _____
Maintain the assigned profile from verified controls.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Operate milling and reclaiming equipment

Worked example · Depth changes across a pass

The approved exercise cut is 40 mm. Safe checks show 40, 41 and then 54 mm near a changed reference surface.

Record each location, notify the lead and hold the affected operation for verification of reference and controls. The 54 mm result should not be concealed by an average. Resume with authorized settings after the cause and required correction are resolved.

Describe a safe supervised sequence for forward–stop–reverse–stop on a flat training pad.

Describe a safe supervised sequence for forward–stop–reverse–stop on a flat training pad.

Answer guidance.

Precheck and brief; confirm clear route and signals; use the OMM travel controls; move smoothly; stop; recheck the reverse route and acknowledgement; reverse and stop; apply the specified parking/isolation controls. Exact controls are machine-specific.

Observe: Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

A 75 m pass takes 15 min of actual travel

A 75 m pass takes 15 min of actual travel. Find average travel speed in m/min; explain why this is not a recommended setting.

Answer guidance.

75/15 = 5 m/min. It is a recorded average, not a safe or productive setting for every material, depth, machine or site.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Design a one-pass log that helps the next operator understand a sensor-reference change.

Design a one-pass log that helps the next operator understand a sensor-reference change.

Answer guidance.

Include machine, station range, depth target/readings, reference type/change, settings as recorded, time, observed response, stop/action and authorization to resume.

Observe: Writing, Digital, Collaboration, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Precheck, start, monitor and shut down using the matching OMM.	0 / 1 / 2	Evidence/help: _____
Move forward/stop/reverse/stop on flat ground and apply parking controls.	0 / 1 / 2	Evidence/help: _____
Raise/lower/swing conveyor only within the cleared envelope.	0 / 1 / 2	Evidence/help: _____
Set and verify depth; follow the line during an approved supervised cut.	0 / 1 / 2	Evidence/help: _____
Coordinate with ground crew and haul driver or approved side-discharge plan.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Follow shut down procedures

Worked example · Service access tomorrow

The nearest parking place is level but blocks an emergency access route. Another approved area provides stable ground and service access.

Use the approved location that preserves emergency access. Follow the machine shutdown procedure, secure components and parking controls, inspect safely and hand over the final state. Convenience does not justify blocking the route.

Name four items in a useful shutdown handover.

Name four items in a useful shutdown handover.

Answer guidance.

Machine/serial and meter; parking location/state; defects and isolation/out-of-service status; responsible person and next required action. Include keys/access and any residual hazards as applicable.

Observe: Writing, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

A checklist has ten equally weighted scored items and a learner earns 16/20

A checklist has ten equally weighted scored items and a learner earns 16/20. A critical parking check failed. Is the learner released?

Answer guidance.

No. 16/20 is 80%, but every safety-critical item must pass and the unsafe condition must be corrected. The numerical threshold cannot compensate for a critical failure.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

A colleague wants to inspect beneath a raised conveyor after shutdown

A colleague wants to inspect beneath a raised conveyor after shutdown. Explain the missing decision.

Answer guidance.

Determine and apply the required machine-specific isolation, stored-energy control and approved support before access. Engine shutdown alone is insufficient.

Observe: Communication, Reading through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Select a stable parking location preserving access.	0 / 1 / 2	Evidence/help: _____
Complete the actual shutdown and component-support procedure.	0 / 1 / 2	Evidence/help: _____
Inspect safely, secure access and communicate defects.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Transport milling and reclaiming equipment

Worked example · A mass record that excludes attachments

The transport plan lists a base machine mass of 19,800 kg. A fitted conveyor option and attachments add 1,250 kg.

Planned machine load is at least 21,050 kg before any other cargo. Confirm actual configuration, carrier/ramp/axle ratings and route requirements. The base-machine figure is insufficient for release; obtain the transport lead's verified plan.

A carrier's fictional available payload is 22,000 kg

A carrier's fictional available payload is 22,000 kg. Machine plus attachments is 21,050 kg and separate equipment is 1,200 kg. Find the excess.

Answer guidance.

Total 22,250 kg, exceeding available payload by 250 kg. Hold and revise the load plan; also check axle distribution and all other limits.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

List information needed before selecting tie-downs.

List information needed before selecting tie-downs.

Answer guidance.

Actual machine mass/configuration, designated attachment points, restraint arrangement, rated components and condition, applicable Standard 10/BC requirements, carrier points and manufacturer transport instructions.

Observe: Reading, Writing through the learner's actual contribution, calculation, explanation or saved work.

During loading the guide disappears from view

During loading the guide disappears from view. What happens next?

Answer guidance.

Stop safely, maintain the machine’s controlled state, re-establish the guide’s safe position and communication, then resume only after confirmation. Do not continue from memory of the route.

Observe: Communication, Collaboration, Adaptability through the learner’s actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Verify roles, carrier/ramp suitability and transport configuration.	0 / 1 / 2	Evidence/help: _____
Prepare attachments and approved tie-down points.	0 / 1 / 2	Evidence/help: _____
Demonstrate controlled loading preparation and final securement checks within the planned time without rushing safety.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Integrated practice · all nine Skills for Success

Hand over a controlled milling pass

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional approved training strip is 60 m × 2 m with a 0.04 m nominal cut. Safe recorded checks at three locations are 40, 41 and 53 mm. The last location follows a disturbed reference. A draft handover states “average depth acceptable; continue.” No project tolerance is supplied. The discharge route remains within a controlled area.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Evidence and feedback: Use the exact case facts; distinguish observation, assumption and missing authority.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Evidence and feedback: The note preserves verified facts, units and unresolved holds without inventing a release.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Evidence and feedback: Use the worked numerical key below; show the relationship and its practical limits.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Evidence and feedback: Record the product/capabilities, prompt, source facts, saved files and checks. A chat response alone is not a created or verified file. Check formulas and preserve originals.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Evidence and feedback: Separate the observed issue from a guessed cause; preserve evidence and identify the correct decision role.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Evidence and feedback: Observe an actual exchange and confirmation of the essential facts; do not infer communication from a shared file.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Evidence and feedback: Both learners take an identifiable turn; feedback is respectful and tied to the case evidence.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Evidence and feedback: Use the change-response key below; retain the earlier version and do not alter unrelated facts.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Evidence and feedback: The visual/media has accurate labels, source credits and a stated purpose. Inspect generated diagrams/media; do not invent technical settings. Keep a transcript/captions when producing audio/video.

Change card · Use after the first attempt

The next shift uses a different machine model and a different rotor configuration.

Verified case key

Geometric removed volume = 4.8 m³ in place. Mean of readings = 44.67 mm, but the mean cannot establish conformity or hide the 53 mm reading. Hold the affected work and verify the reference, limits and authorized correction.

Change-response key

Reconfirm matching OMM, controls, maintenance and setup; transfer the verified site information without transferring unsupported machine settings.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person's actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.